

A crystalline material is one in which the atoms are situated in a repeating and periodic array over large atomic distances.

Susmita Panday Shuvoshree

2021-22, MSE, RU

Lattice means a three dimensional array of points ~~point~~ coinciding with atom positions.

Basis means the arrangement of atoms associated with each lattice point.

- When lattice is replaced by atoms, molecules or ions, it will be basis.
- There is no atom in lattice
- Lattice is regular and periodic arrangement of points in space.

Crystal $\rightarrow$ **lattice + basis**

A unit cell is the basic structural unit or building block of the crystal structure and defines the crystal structure by virtue of its geometry and the atom positions within.

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Bravais lattice

Bravais lattice refers to the 14 different 3 dimensional configurations into which atoms can be arranged in a unit cell of crystals.

Lattice parameters

A unit cell is completely defined in terms of six parameters: the three edge lengths a, b, c and the three interaxial angles α, β, γ . These are termed as lattice parameters.

$a, b, c \rightarrow$ axial lengths

$\alpha, \beta, \gamma \rightarrow$ interaxial angles

Interaxial length between x and $y \rightarrow \gamma$

" " " y and $z \rightarrow \alpha$

" " " z and $x \rightarrow \beta$

Crystal system	Axial Relationships	Interaxial angles
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Cubic	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$
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Hexagonal	$a=b \neq c$	$\alpha=\beta=90^\circ, \gamma=120^\circ$
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Tetragonal	$a=b \neq c$	$\alpha=\beta=\gamma=90^\circ$
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Rhombohedral	$a=b=c$	$\alpha=\beta=\gamma \neq 90^\circ$
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Orthorhombic	$a \neq b \neq c$	$\alpha=\beta=\gamma=90^\circ$
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Monoclinic	$a \neq b \neq c$	$\alpha=\gamma=90^\circ \neq \beta$
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Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$
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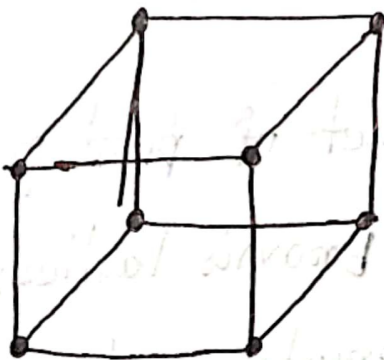
The primitive cell is a unit cell corresponding to a single lattice point.

Bravais lattice of cube

1. Simple cube (SC)
2. Body centered cubic (BCC)
3. Face centered cubic (FCC)

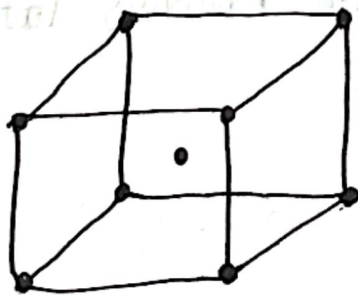
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1. SC



equivalent number of atoms
 $= 8 \times \frac{1}{8} = 1$

BCC

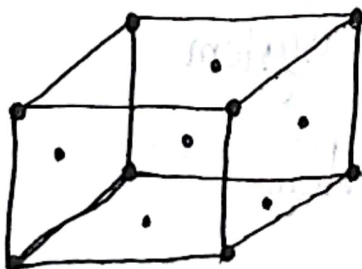


equivalent number of atoms

$$8 \times \frac{1}{8} = 1$$

$$1 + 1 = 2$$

FCC



equivalent number of atoms

$$8 \times \frac{1}{8} = 1, 6 \times \frac{1}{2} = 3$$

$$1 + 3 = 4$$

crystal structure

crystal structure is a description of the orderly arrangement of atoms, ions, or molecules in a crystalline material.

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crystal system

A crystal system is a set of point groups. A lattice system is a set of Bravais lattices. Space groups are classified into crystal systems according to their point groups, and into lattice systems according to their Bravais lattice.

crystal system

1. Cubic system
2. Triclinic system
3. Orthorhombic system
4. Monoclinic system

5. Hexagonal system
6. Tetragonal system
7. Rhombohedral / trigonal system

Lattice site

In crystal lattice, each of atoms/molecules/ions is represented by a single point, those points are called lattice site/point.

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Interstitial site

Interstitial sites are the empty space that exists between the packing of atom in the crystal structure.

The coordination number

For metals, each atom has the same number of nearest neighbor or touching atoms, which is the coordination number.

Atomic packing factor (APF)

The APF is the sum of the sphere volume of all atoms within a unit cell divided by the unit cell volume that is,

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}}$$

Coordination number of

$$\text{SC} = 6$$

$$\text{BCC} = 8$$

$$\text{FCC} = 12$$

$$\text{HCP} = 12$$

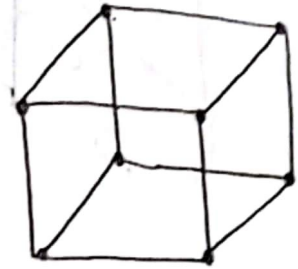
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Relation between atomic radius with axial length / lattice parameter

Date

Simple cubic

The radius of the constituent particle of a simple cubic unit cell is equal to half of the edge length of the unit cell i.e., $r = \frac{1}{2}a$



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Body centered cubic

Here,

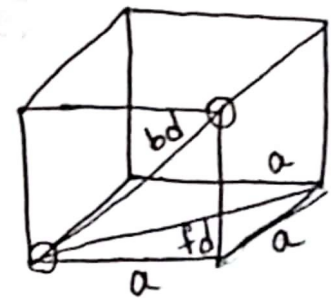
$$fd^2 + a^2 = bd^2$$

$$\Rightarrow 2a^2 + a^2 = (4R)^2$$

$$\Rightarrow 3a^2 = (4R)^2$$

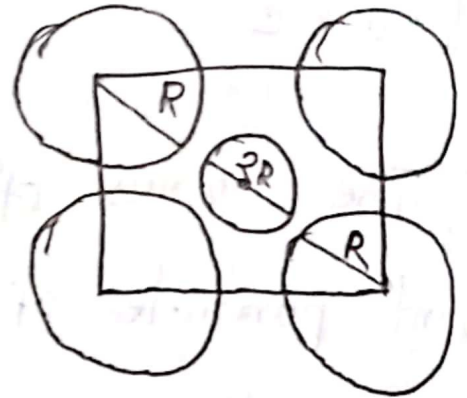
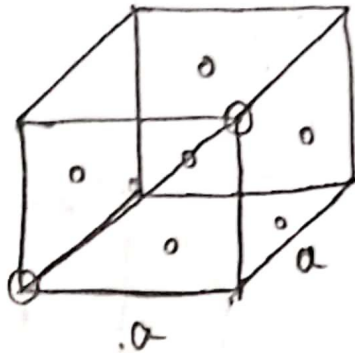
$$\Rightarrow \sqrt{3}a = 4R$$

$$\Rightarrow a = \frac{4R}{\sqrt{3}}$$



$$fd^2 = a^2 + a^2 = 2a^2$$

Face centered cubic



Here,

$$a^2 + a^2 = (4R)^2$$

$$\Rightarrow 2a^2 = (4R)^2$$

$$\Rightarrow \sqrt{2} a = 4R$$

$$\Rightarrow a = \frac{4R}{\sqrt{2}}$$

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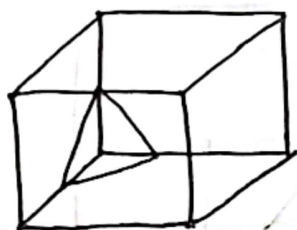
Miller indices

Miller indices are three smallest possible integers h, k, l which have the same ratio as the reciprocals of intercepts of the concerned plane on three axes. The miller indices for a particular plane are determined in the following way.

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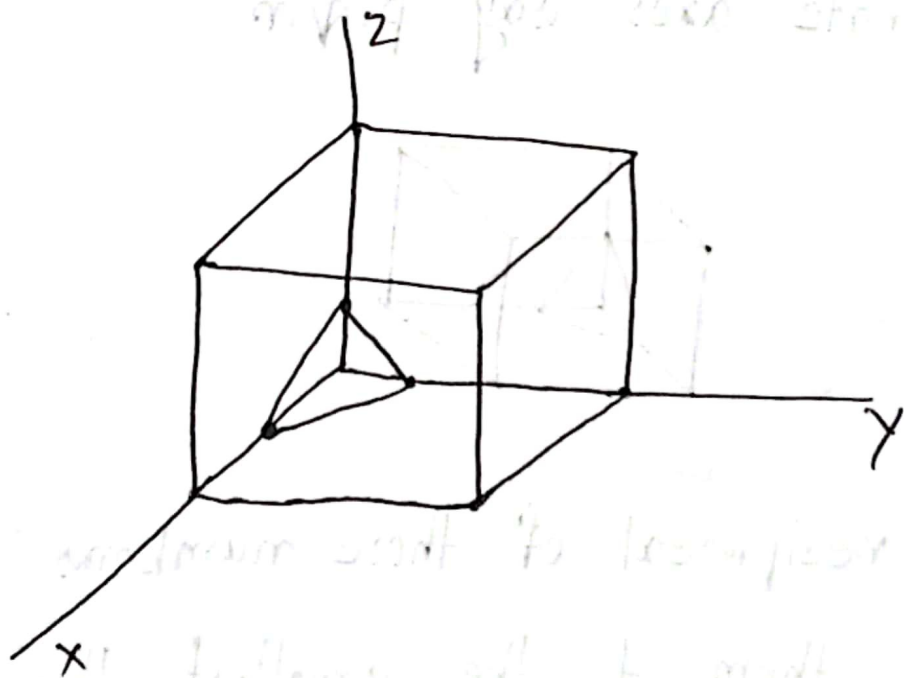
1. First determine the intercepts of the plane of the three co ordinate axes say p, q, r



2. Take the reciprocal of these numbers and by taking LCM volume them to the smallest three intercepts as h, k, l such that $h:k:l = \frac{1}{p} : \frac{1}{q} : \frac{1}{r}$. The result is then enclosed in parenthesis as (h, k, l)

3. These the concerned plane is called to have the Miller indices (hkl)

As an example, for a plane whose intercepts are 2, 3, 4 along these axes. The reciprocals are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. Hence according to the definition of Miller indices $h:k:l = \frac{1}{2}:\frac{1}{3}:\frac{1}{4} = 6:4:3$. Thus the Miller indices are $(624\ 3)$



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$(121) \rightarrow$ a specific plane

$\{100\} \rightarrow$ a set of planes

$[121] \rightarrow$ a specific direction

$\langle 121 \rangle \rightarrow$ a set of directions

Density

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1. Linear density (LD) (cm^{-1})
2. Planar density (PD) (cm^{-2})
3. Volume density (VD) (cm^{-3})

Linear density (LD)

$$P_L = \frac{\text{equivalent number of atoms in direction}}{\text{length in the direction}}$$

Planar density (PD)

$$\rho_p = \frac{\text{equivalent number of atoms in plane}}{\text{area of the plane}}$$

Volume density (VD)

$$\rho_v = \frac{\text{Mass/unit cell}}{\text{volume/unit cell}}$$

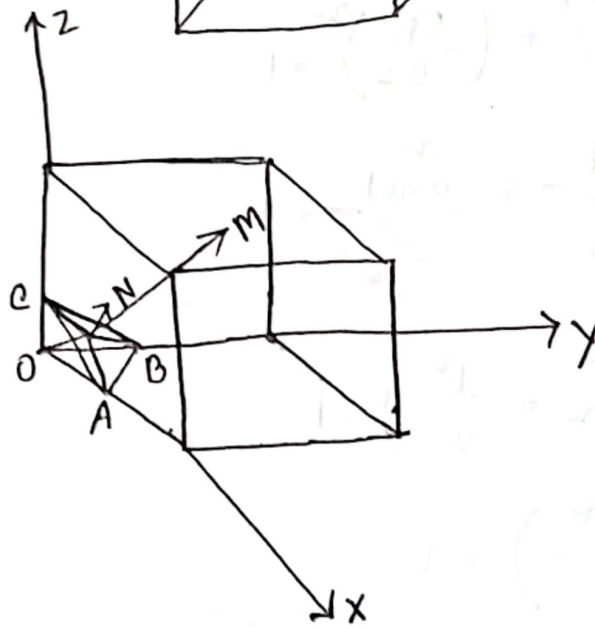
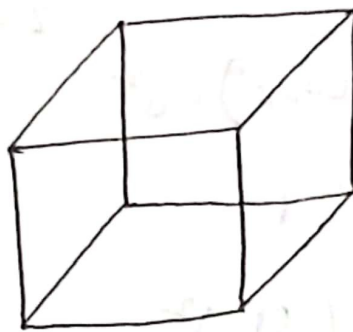
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Interplanar distance/spacing

Derivation of interplanar distance in a cubic lattice.

On

Derive the equation for interplanar distance $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$



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Here,

let,

Interaxial angles α', β', γ'

$$OA = \frac{a}{h}, \cos \alpha' = \frac{ON}{OA}$$

$$OB = \frac{a}{k}, \cos \beta' = \frac{ON}{OB}$$

Here,

$$OA = \frac{a}{h}$$

$$OB = \frac{a}{k}$$

$$OC = \frac{a}{l}$$

$$OC = \frac{a}{1}, \cos \gamma' = \frac{ON}{OC}$$

Here,

$$ON = d_{hkl}$$

$$\Rightarrow d = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow d^2 = x^2 + y^2 + z^2$$

let,

$$x = d \cos \alpha'$$

$$y = d \cos \beta'$$

$$z = d \cos \gamma'$$

$$\Rightarrow d^2 (\cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma') = d^2$$

$$\Rightarrow \cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma' = 1$$

$$\Rightarrow \left(\frac{ON}{OA} \right)^2 + \left(\frac{ON}{OB} \right)^2 + \left(\frac{ON}{OC} \right)^2 = 1$$

$$\Rightarrow \frac{d_{hkl}^2}{\frac{a^2}{h^2}} + \frac{d_{hkl}^2}{\frac{a^2}{k^2}} + \frac{d_{hkl}^2}{\frac{a^2}{l^2}} = 1$$

$$\Rightarrow d_{hkl}^2 \left(\frac{h^2}{a^2} + \frac{k^2}{a^2} + \frac{l^2}{a^2} \right) = 1$$

$$\Rightarrow d_{hkl}^2 \left(\frac{h^2 + k^2 + l^2}{a^2} \right) = 1$$

$$\Rightarrow d_{hkl}^2 = \frac{a^2}{h^2 + k^2 + l^2}$$

$$\Rightarrow d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

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#

1. (h, k, l)

2. (hkl)

3. $\{hkl\}$

4. $[hkl]$

5. $\langle hkl \rangle$

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If the planar density is same for all plane then a set of planes can be determined.

$$\{100\} \rightarrow (100) (001) (010) (\bar{1}00) (0\bar{1}0) (00\bar{1})$$

APF of SC:

$$\text{volume of atom} = \frac{4}{3} \pi R^3$$

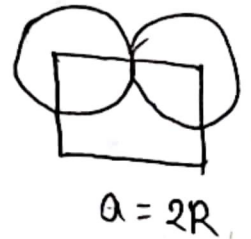
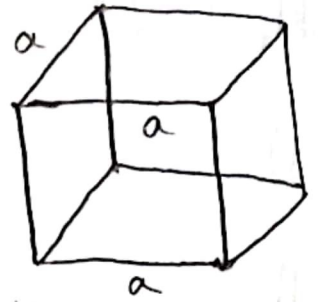
$$\text{volume of unit cell} = a^3$$

$$= (2R)^3$$

$$= 8R^3$$

$$\therefore \text{APF} = \frac{\frac{4}{3} \pi R^3}{8R^3}$$

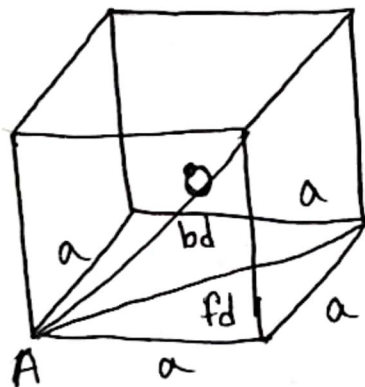
$$= 0.52 = 52\%$$



APF of BCC

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$$\text{Number of equivalent atoms in 1 unit cell} = \left(8 \times \frac{1}{8}\right)$$
$$= 2$$



derivation

$$\left[a = \frac{4R}{\sqrt{3}} \right] \rightarrow$$

$$\text{volume of atoms} = 2 \times \frac{4}{3} \pi R^3$$

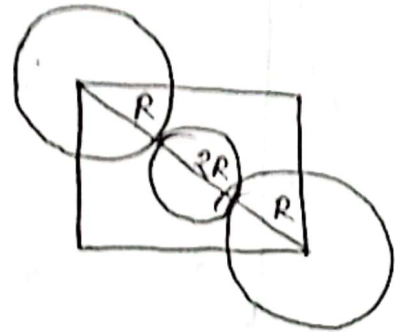
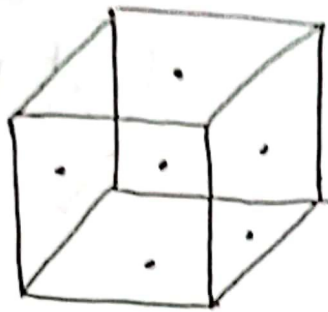
$$\text{volume of unit cell} = a^3 = \left(\frac{4R}{\sqrt{3}} \right)^3$$

$$APF = \frac{2 \times \frac{4}{3} \pi R^3}{\left(\frac{4R}{\sqrt{2}}\right)^3}$$

$$= 0.68 = 68\%$$

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APF of FCC



$$\begin{aligned} \text{number of equivalent atoms} &= \left(8 \times \frac{1}{8}\right) + \left(6 \times \frac{1}{2}\right) \\ &= 4 \end{aligned}$$

$$\text{volume of atom} = 4 \times \frac{4}{3} \pi R^3$$

$$\text{volume of unit cell} = a^3 = \left(\frac{4R}{\sqrt{2}}\right)^3$$

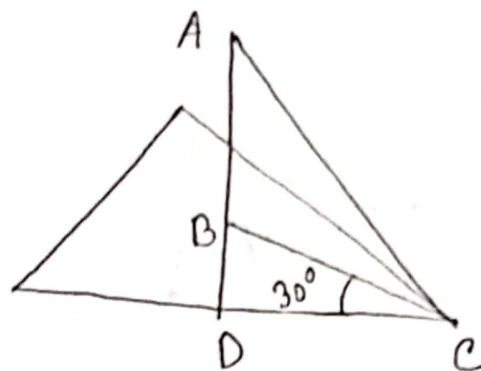
derivation
 $\left(a = \frac{4R}{\sqrt{2}}\right) \uparrow$

$$\therefore APF = \frac{4 \times \frac{4}{3} \pi R^3}{\left(\frac{4R}{\sqrt{2}}\right)^3}$$

$$= 0.74 = 74\%$$

APF of HCP

Number of equivalent atoms
in unit cell = $2(6 \times \frac{1}{6}) + (2 \times \frac{1}{2}) + 3$
 $= 6$



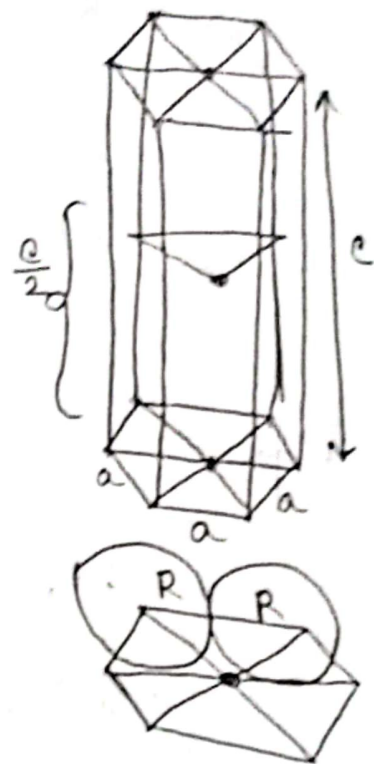
In $\triangle BCD$ triangle,

$$\cos 30^\circ = \frac{CD}{BC}$$

$$\Rightarrow BC = \frac{CD}{\cos 30^\circ}$$

$$\Rightarrow BC = \frac{R}{\frac{\sqrt{3}}{2}}$$

$$\therefore BC = \frac{2R}{\sqrt{3}} = \frac{2 \times \frac{a}{2}}{\sqrt{3}} = \frac{a}{\sqrt{3}}$$



$$a = 2R$$

$$\Rightarrow R = \frac{a}{2}$$

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4 In $\triangle ABC$ triangle

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow a^2 = \frac{c^2}{4} + \frac{a^2}{3}$$

$$\Rightarrow \frac{c^2}{4} = a^2 - \frac{a^2}{3} = \frac{2a^2}{3}$$

$$\Rightarrow \frac{a^2}{c^2} = \frac{3}{8}$$

$$\Rightarrow \frac{a}{c} = \sqrt{\frac{3}{8}}$$

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$\therefore$ Total number of HCP = Area of the base $\times$ height

$$= 6 \times \frac{\sqrt{3}}{4} \times a^2 \times c$$

$$= \frac{3\sqrt{3}}{2} a^2 c$$

$$\begin{aligned}\therefore \text{APF}_{\text{HCP}} &= \frac{6 \times \frac{4}{3} \pi R^3}{\frac{3\sqrt{3}}{2} a^2 c} \\ &= \frac{8\pi \left(\frac{a}{2}\right)^3 \times 2}{3\sqrt{3} a^2 c} \\ &= \frac{16\pi a^3}{24\sqrt{3} a^2 c}\end{aligned}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{a}{c}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{8}}$$

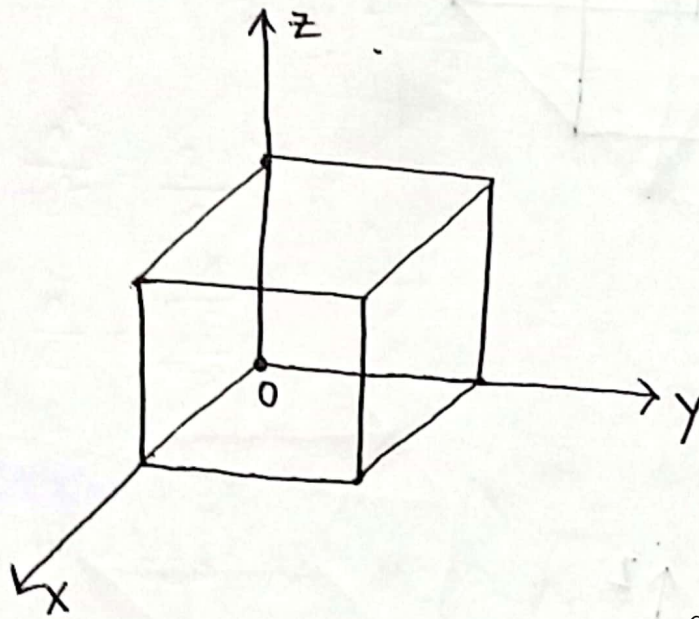
$$= 0.74 = 74\%$$

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plane and direction

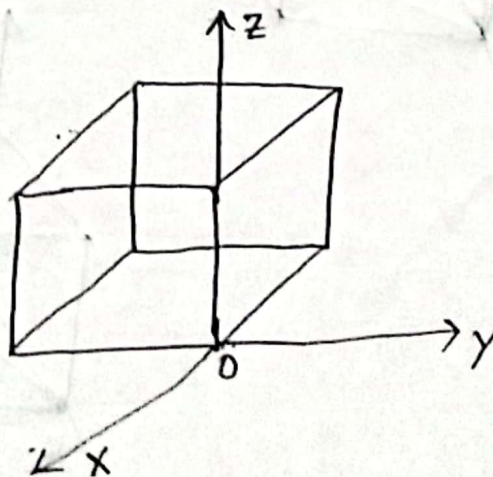
If

1. three of the axes are (+)ve ($x y z$)

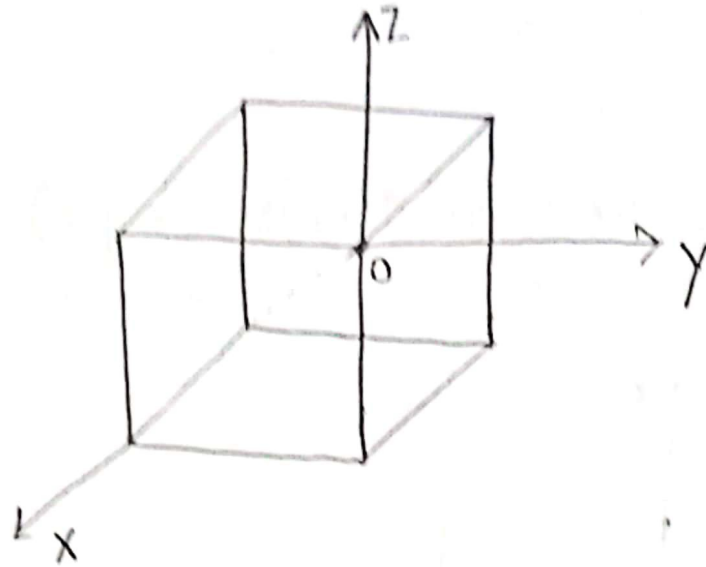


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2. two of the axes are (-)ve ($\bar{x} \bar{y} z$)



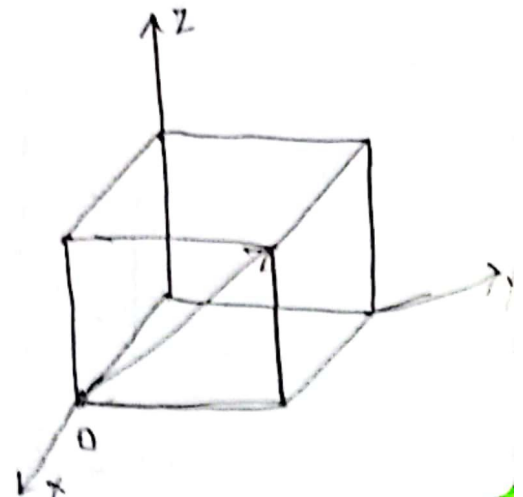
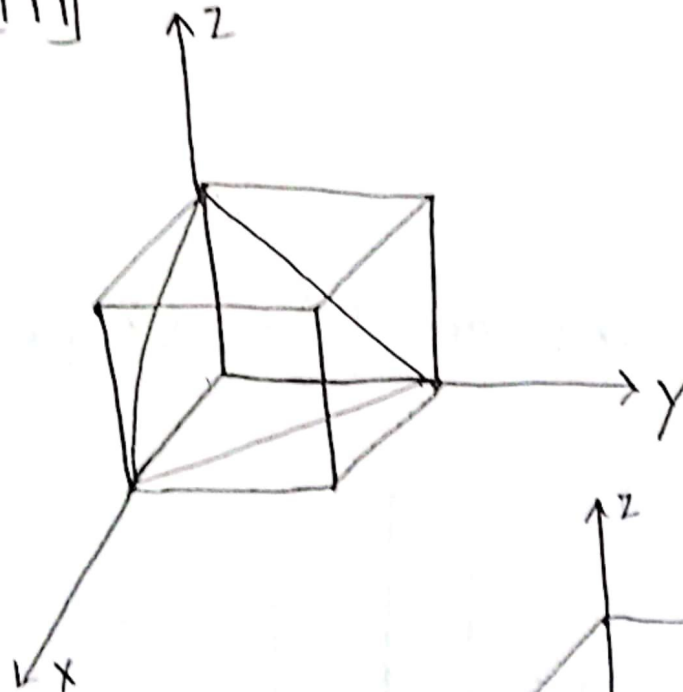
3. three of the axes are (-ve $\bar{x} \bar{y} \bar{z}$)

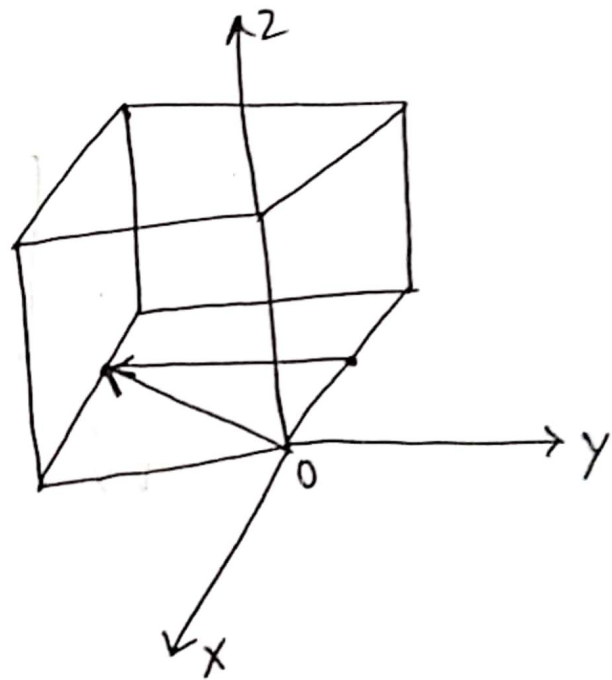
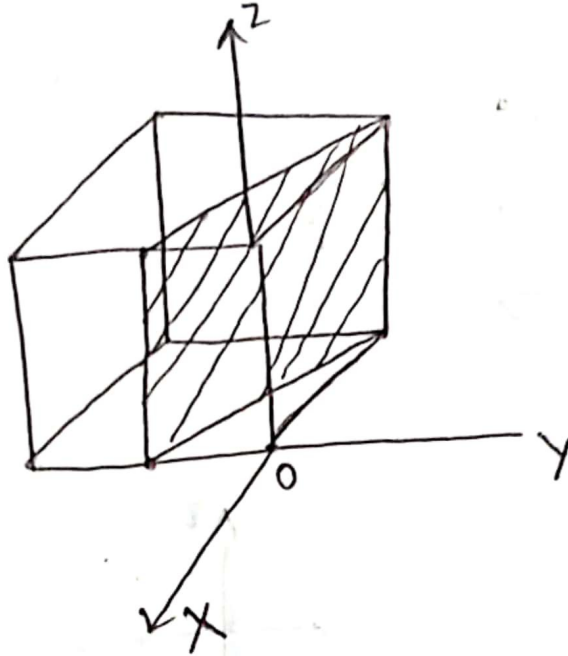


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Plane (111)

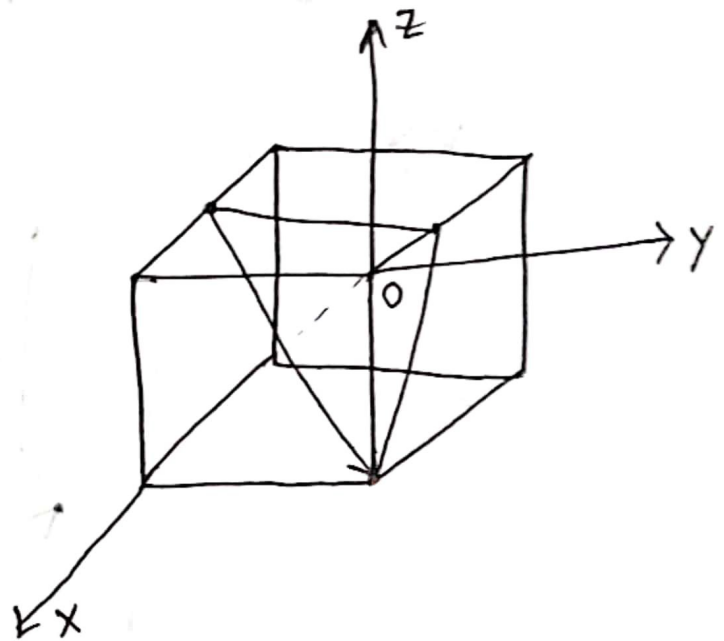
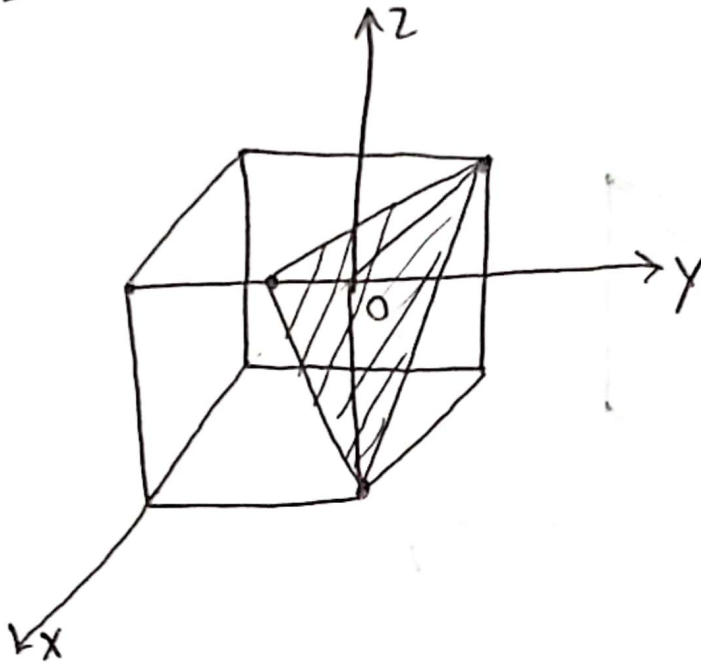
Direction $[h \ k \ l]$



plane $(\bar{1} \bar{2} 0)$ Direction $[\bar{1} \bar{2} 0]$ Susmita Panday Shuvoshree
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plane $(\bar{1} \bar{2} \bar{1})$

Direction $[\bar{1} \bar{2} \bar{1}]$



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